

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 07 • Sequences and Series

With Answers

Questions with full worked solutions

- 24 pages
- Past papers up to January 2026

PURE 2 | TOPIC 07

Sequences & Series

Your classified route

10

QUESTIONS

68

MARKS

2

FAMILIES

2

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 07 | FAMILY 01

Unknown Count

WHAT THIS FAMILY ASKS

Recognise the unknown count demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**4 classified questions**

29 total marks

T07

07_MIX

07_MIX_CORE

CHAPTER

Sequences & Series

FAMILY

Unknown count

VARIANT

core family move

7 marks

WMA12/01 JANUARY 2021

Question 8

Past-paper question

WMA12-01_2021_Jan_Q08

8. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$
$$a_1 = p - 3$$

where p is a constant.

- (a) Find an expression for a_2 in terms of p , giving your answer in simplest form.

(1)

Given that $\sum_{n=1}^3 a_n = p + 15$

- (b) find the possible values of a_2

(6)

T07 Sequences & Series

07_MIX Unknown count

07_MIX_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 January 2021

WMA12-01_2021_Jan_Q08

1

Find a two

Since $a_1 = p - 3$, we have $a_1 + 3 = p$. Therefore $a_2 = 2(a_1 + 3)^2 - 7 = 2p^2 - 7$.

2

Find a three

$a_2 + 3 = 2p^2 - 4$, so $a_3 = 2(2p^2 - 4)^2 - 7 = 8(p^2 - 2)^2 - 7$.

3

Use the given sum

$a_1 + a_2 + a_3 = p + 15$.

4

Substitute the expressions

$(p - 3) + (2p^2 - 7) + 8(p^2 - 2)^2 - 7 = p + 15$.

5

Simplify

Cancelling p and expanding gives $8p^4 - 30p^2 = 0$, so $2p^2(4p^2 - 15) = 0$.

6

Find the possible a two values

If $p = 0$, then $a_2 = -7$. If $p^2 = \frac{15}{4}$, then $a_2 = 2 \cdot \frac{15}{4} - 7 = \frac{1}{2}$.

FINAL ANSWER

WMA12-01_2021_Jan_Q08

(a) $a_2 = 2p^2 - 7$. (b) $a_2 = -7$ or $\frac{1}{2}$.

T07

07_MIX

07_MIX_CORE

CHAPTER

Sequences & Series

FAMILY

Unknown count

VARIANT

core family move

5 marks

WMA12/01 OCTOBER 2021

Question 2

Past-paper question

WMA12-01_2021_Oct_Q02

2. A sequence is defined by

$$u_1 = 6$$
$$u_{n+1} = ku_n + 3$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for u_3 (2)

Given that $\sum_{n=1}^3 u_n = 117$

- (b) find the value of k . (3)

T07 Sequences & Series

07_MIX Unknown count

07_MIX_CORE core family
move

WORKED SOLUTION

Question 2

WMA12/01 October 2021

WMA12-01_2021_Oct_Q02

1

Find u two

$$u_2 = ku_1 + 3 = 6k + 3.$$

2

Find u three

$$u_3 = ku_2 + 3 = k(6k + 3) + 3 = 6k^2 + 3k + 3.$$

3

Use the given sum

$$u_1 + u_2 + u_3 = 117, \text{ so } 6 + (6k + 3) + (6k^2 + 3k + 3) = 117.$$

4

Simplify the equation

$$6k^2 + 9k + 12 = 117, \text{ hence } 2k^2 + 3k - 35 = 0.$$

5

Solve for k

$$(2k - 7)(k + 5) = 0. \text{ Since } k \text{ is positive, } k = \frac{7}{2}.$$

FINAL ANSWER

WMA12-01_2021_Oct_Q02

$$(a) u_3 = 6k^2 + 3k + 3. \quad (b) k = \frac{7}{2}.$$

T07

07_MIX

07_MIX_CORE

CHAPTER

Sequences & Series

FAMILY

Unknown count

VARIANT

core family move

8 marks

WMA12/01 MAY/JUNE 2023

Question 11

Past-paper question

WMA12-01_2023_MayJune_Q11

11. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where a and b are constants.

(a) Find, in terms of a and b ,

(i) u_2

(ii) u_3

(2)

Given

- $\sum_{n=1}^3 u_n = 153$

- $b = a + 9$

(b) show that

$$a^2 - 5a - 66 = 0$$

(3)

(c) Hence find the larger possible value of u_2

(3)

T07 Sequences & Series

07_MIX Unknown count

07_MIX_CORE core family
move

WORKED SOLUTION

Question 11

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q11

1

Find u_2 Using $u_{n+1} = b - au_n$ and $u_1 = 3$, we get $u_2 = b - a(3) = b - 3a$.

2

Find u_3 $u_3 = b - au_2 = b - a(b - 3a) = b - ab + 3a^2$.

3

Use the sum condition $u_1 + u_2 + u_3 = 153$, so $3 + (b - 3a) + (b - ab + 3a^2) = 153$.

4

Substitute b equals a plus nineUsing $b = a + 9$, the left side becomes $3 + (a + 9) - 3a + (a + 9) - a(a + 9) + 3a^2$.

5

SimplifyThis is $2a^2 - 10a + 21$, so $2a^2 - 10a + 21 = 153$.

6

Show the required equation $2a^2 - 10a - 132 = 0$. Dividing by 2 gives $a^2 - 5a - 66 = 0$.

7

Solve for a $a^2 - 5a - 66 = (a - 11)(a + 6) = 0$, so $a = 11$ or $a = -6$.

8

Find the corresponding u_2 valuesIf $a = 11$, then $b = 20$ and $u_2 = 20 - 3(11) = -13$. If $a = -6$, then $b = 3$ and $u_2 = 3 - 3(-6) = 21$.

9

Choose the larger valueThe larger possible value of u_2 is 21.**FINAL ANSWER**

WMA12-01_2023_MayJune_Q11

(a)(i) $u_2 = b - 3a$. (a)(ii) $u_3 = b - ab + 3a^2$. (b) **Proven.** (c) 21.

T07

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07_MIX_CORE

CHAPTER

Sequences & Series

FAMILY

Unknown count

VARIANT

core family move

9 marks

WMA12/01 MAY/JUNE 2025

Question 7

Past-paper question

WMA12-01_2025_MayJune_Q07

- 7: **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) A **geometric** series begins

$$10 + 8 + 6.4 + \dots$$

- (a) Find the sum to infinity of this series.

(2)

Given that the k th term of this series is less than 0.0005

- (b) use algebra to find the smallest possible value of k .

(3)

- (ii) An **arithmetic** series begins

$$850 + 843 + 836 + \dots$$

Given that the sum of the first n terms of this series is S_n
find the greatest possible value of S_n

(4)

T07 Sequences & Series

07_MIX Unknown count

07_MIX_CORE core family
move

WORKED SOLUTION

Question 7

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q07

1

Identify the geometric sequenceThe first term is 10 and the common ratio is $\frac{8}{10} = 0.8$.

2

Find the sum to infinity

$$S_{\infty} = \frac{10}{1-0.8} = 50.$$

3

Write the k th termThe k th term is $10(0.8)^{k-1}$.

4

Use the inequality

$$10(0.8)^{k-1} < 0.0005, \text{ so } (0.8)^{k-1} < 0.00005.$$

5

Solve with logarithms

Taking logarithms gives $(k-1)\log(0.8) < \log(0.00005)$. Since $\log(0.8) < 0$, $k-1 > \frac{\log(0.00005)}{\log(0.8)} = 44.381 \dots$

6

Find the smallest integerThus $k > 45.381 \dots$, so the smallest possible value is $k = 46$.

7

Write the arithmetic sumFor $850 + 843 + 836 + \dots$, the first term is 850 and the common difference is -7 .

8

Form S_n

$$S_n = \frac{n}{2}\{2(850) + (n-1)(-7)\} = \frac{n}{2}(1707 - 7n).$$

9

Find the best integer n This quadratic has its maximum near $n = \frac{1707}{14} = 121.928 \dots$, so check $n = 122$.

10

Find the greatest sum

$$S_{122} = \frac{122}{2}(1707 - 854) = 61(853) = 52033.$$

FINAL ANSWER

WMA12-01_2025_MayJune_Q07

(i)(a) 50. (i)(b) 46. (ii) 52033.

DR ESLAM AHMED

TOPIC 07 | FAMILY 02

Sigma Sums

WHAT THIS FAMILY ASKS

Recognise the sigma sums demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

6 classified questions

39 total marks

T07

07_SIGMA

07_SIGMA_CORE

CHAPTER

Sequences & Series

FAMILY

Sigma sums

VARIANT

core family move

7 marks

WMA12/01 JANUARY 2020

Question 8

Past-paper question

WMA12-01_2020_Jan_Q08

8. (i) An arithmetic series has first term a and common difference d .

Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a + (n-1)d\} \quad (3)$$

- (ii) A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_n = 5n + 3(-1)^n$$

Find the value of

(a) u_5 (1)

(b) $\sum_{n=1}^{59} u_n$ (3)

T07 Sequences & Series

07_SIGMA Sigma sums

07_SIGMA_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 January 2020

WMA12-01_2020_Jan_Q08

1

Write the series forwards

$$S_n = a + (a + d) + (a + 2d) + \cdots + (a + (n - 1)d).$$

2

Write the series backwards

$$S_n = (a + (n - 1)d) + (a + (n - 2)d) + \cdots + a.$$

3

Add the two forms

Each pair adds to $2a + (n - 1)d$, and there are n pairs. Hence $2S_n = n\{2a + (n - 1)d\}$.

4

Complete the proof

Dividing by 2 gives $S_n = \frac{n}{2}\{2a + (n - 1)d\}$.

5

Find u_5

$$u_5 = 5(5) + 3(-1)^5 = 25 - 3 = 22.$$

6

Sum the sequence

$$\sum_{n=1}^{59} u_n = 5 \sum_{n=1}^{59} n + 3 \sum_{n=1}^{59} (-1)^n.$$

7

Calculate the total

$5 \sum_{n=1}^{59} n = 5 \cdot \frac{59 \cdot 60}{2} = 8850$, and $\sum_{n=1}^{59} (-1)^n = -1$. Therefore the total is $8850 - 3 = 8847$.

FINAL ANSWER

WMA12-01_2020_Jan_Q08

(i) **Proven.** (ii) (a) 22. (ii) (b) 8847.

T07

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CHAPTER

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core family move

7 marks

WMA12/01 OCTOBER 2022

Question 3

Past-paper question

WMA12-01_2022_Oct_Q03

3. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(a) (i) a_1

(ii) a_2

(iii) a_3

(3)

- (b) Hence find the exact value of

$$\sum_{n=1}^{50} \left\{ n + \cos^2\left(\frac{n\pi}{3}\right) \right\}$$

You must make your method clear.

(4)

T07 Sequences & Series

07_SIGMA Sigma sums

07_SIGMA_CORE core family
move

WORKED SOLUTION

Question 3

WMA12/01 October 2022

WMA12-01_2022_Oct_Q03

1

Find the first term

$$a_1 = \cos^2\left(\frac{\pi}{3}\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

2

Find the second term

$$a_2 = \cos^2\left(\frac{2\pi}{3}\right) = \left(-\frac{1}{2}\right)^2 = \frac{1}{4}.$$

3

Find the third term

$$a_3 = \cos^2(\pi) = (-1)^2 = 1.$$

4

Recognise the repeating pattern

The values of $\cos^2\left(\frac{n\pi}{3}\right)$ repeat every 3 terms as $\frac{1}{4}, \frac{1}{4}, 1$.

5

Sum the n part

$$\sum_{n=1}^{50} n = \frac{50 \cdot 51}{2} = 1275.$$

6

Sum the cosine part

There are 16 complete groups of 3 terms in the first 48 terms, then the first 2 terms repeat. Hence the cosine-square sum is $16\left(\frac{1}{4} + \frac{1}{4} + 1\right) + \frac{1}{4} + \frac{1}{4} = 24 + \frac{1}{2} = \frac{49}{2}$.

7

Add the two sums

$$\sum_{n=1}^{50} \left(n + \cos^2\left(\frac{n\pi}{3}\right)\right) = 1275 + \frac{49}{2} = \frac{2599}{2}.$$

FINAL ANSWER

WMA12-01_2022_Oct_Q03

$$(a) a_1 = \frac{1}{4}, a_2 = \frac{1}{4}, a_3 = 1. \quad (b) \frac{2599}{2}.$$

T07

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5 marks

WMA12/01 OCTOBER 2023

Question 2

Past-paper question

WMA12-01_2023_Oct_Q02

2. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$

$$u_{n+1} = 2 - \frac{4}{u_n}$$

- (a) Find the value of u_2 , the value of u_3 and the value of u_4

(3)

- (b) Find the value of

$$\sum_{r=1}^{100} u_r$$

(2)

T07 Sequences & Series

07_SIGMA Sigma sums

07_SIGMA_CORE core family
move

WORKED SOLUTION

Question 2

WMA12/01 October 2023

WMA12-01_2023_Oct_Q02

1

Find u_2

$$u_2 = 2 - \frac{4}{u_1} = 2 - \frac{4}{3} = \frac{2}{3}.$$

2

Find u_3

$$u_3 = 2 - \frac{4}{u_2} = 2 - \frac{4}{2/3} = 2 - 6 = -4.$$

3

Find u_4

$$u_4 = 2 - \frac{4}{u_3} = 2 - \frac{4}{-4} = 3.$$

4

Notice the cycle

The sequence starts $3, \frac{2}{3}, -4, 3, \frac{2}{3}, -4, \dots$, so it repeats every 3 terms.

5

Sum one cycle

$$3 + \frac{2}{3} - 4 = -\frac{1}{3}.$$

6

Use 100 terms

There are 33 complete cycles in the first 99 terms, then one extra term $u_{100} = 3$.

7

Find the sum

$$\sum_{r=1}^{100} u_r = 33 \left(-\frac{1}{3} \right) + 3 = -11 + 3 = -8.$$

FINAL ANSWER

WMA12-01_2023_Oct_Q02

(a) $u_2 = \frac{2}{3}$, $u_3 = -4$, $u_4 = 3$. (b) -8 .

T07

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8 marks

WMA12/01 JANUARY 2024

Question 5

Past-paper question

WMA12-01_2024_Jan_Q05

5. (i) Find the value of

$$\sum_{r=1}^{\infty} 6 \times (0.25)^r \quad (3)$$

- (ii) A sequence $u_1, u_2, u_3, \dots$ is defined by

$$\begin{aligned} u_1 &= 3 \\ u_{n+1} &= \frac{u_n - 3}{u_n - 2} \quad n \in \mathbb{N} \end{aligned}$$

- (a) Show that this sequence is periodic. (2)

- (b) State the order of this sequence. (1)

- (c) Hence find

$$\sum_{n=1}^{70} u_n \quad (2)$$

T07 Sequences & Series

07_SIGMA Sigma sums

07_SIGMA_CORE core family
move

WORKED SOLUTION

Question 5

WMA12/01 January 2024

WMA12-01_2024_Jan_Q05

1

Identify the geometric series $\sum_{r=1}^{\infty} 6(0.25)^r$ has first term $6(0.25) = 1.5$ and common ratio 0.25.

2

Sum to infinity

$$S_{\infty} = \frac{1.5}{1-0.25} = \frac{1.5}{0.75} = 2.$$

3

Find the first few terms

$$u_1 = 3. \text{ Then } u_2 = \frac{3-3}{3-2} = 0.$$

4

Continue the recurrence

$$u_3 = \frac{0-3}{0-2} = \frac{3}{2}, \text{ and } u_4 = \frac{\frac{3}{2}-3}{\frac{3}{2}-2} = 3.$$

5

Show periodicitySince $u_4 = 3 = u_1$, the terms repeat as $3, 0, \frac{3}{2}, 3, 0, \frac{3}{2}, \dots$

6

State the order

The repeating cycle has length 3, so the order is 3.

7

Sum one cycle

$$\text{One full cycle sums to } 3 + 0 + \frac{3}{2} = \frac{9}{2}.$$

8

Use 70 termsSince $70 = 23 \cdot 3 + 1$, the first 70 terms contain 23 complete cycles and one extra term $u_1 = 3$.

9

Find the required sum

$$\sum_{n=1}^{70} u_n = 23 \left(\frac{9}{2} \right) + 3 = \frac{207}{2} + \frac{6}{2} = \frac{213}{2}.$$

FINAL ANSWER

WMA12-01_2024_Jan_Q05

(i) 2. (ii)(a) **Periodic.** (ii)(b) **Order 3.** (ii)(c) $\frac{213}{2}$.

T07

07_SIGMA

07_SIGMA_CORE

CHAPTER

Sequences & Series

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5 marks

WMA12/01 MAY/JUNE (R) 2024

Question 1

Past-paper question

WMA12-01_2024_MayJuneR_Q01

1. The sequence $u_1, u_2, u_3, \dots$ satisfies

$$u_{n+2} = 3u_{n+1} - 2u_n$$

Given that

- $u_1 = 7$
- $u_3 = 4$

- (a) find the value of u_2

(2)

- (b) find $\sum_{r=1}^4 (u_r + 2r)$

(3)

T07 Sequences & Series

07_SIGMA Sigma sums

07_SIGMA_CORE core family
move

WORKED SOLUTION

Question 1

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q01

1

Use the recurrence at n equals one

$$u_3 = 3u_2 - 2u_1.$$

2

Substitute given values

$$4 = 3u_2 - 2(7), \text{ so } 4 = 3u_2 - 14.$$

3

Find u_2

$$3u_2 = 18, \text{ hence } u_2 = 6.$$

4

Find u_4

$$u_4 = 3u_3 - 2u_2 = 3(4) - 2(6) = 0.$$

5

Evaluate the sum

$$\sum_{r=1}^4 (u_r + 2r) = (7 + 2) + (6 + 4) + (4 + 6) + (0 + 8).$$

6

Add the terms

$$\text{The sum is } 9 + 10 + 10 + 8 = 37.$$

FINAL ANSWER

$$(a) u_2 = 6. \quad (b) 37.$$

WMA12-01_2024_MayJuneR_Q01

T07

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CHAPTER

Sequences & Series

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7 marks

WMA12/01 OCTOBER 2024

Question 2

Past-paper question

WMA12-01_2024_Oct_Q02

2. A sequence of numbers $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where k is a constant.

- (a) Show that $u_5 = 7$

(3)

Given that $\sum_{r=1}^4 u_r = 30$

- (b) find the value of k .

(2)

- (c) Hence find the value of $\sum_{r=1}^{150} u_r$

(2)

T07 Sequences & Series

07_SIGMA Sigma sums

07_SIGMA_CORE core family
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WORKED SOLUTION

Question 2

WMA12/01 October 2024

WMA12-01_2024_Oct_Q02

1

Find the first few terms $u_1 = 7, u_2 = (-1)^1 u_1 + k = k - 7, u_3 = (-1)^2 u_2 + k = 2k - 7, \text{ and } u_4 = (-1)^3 u_3 + k = 7 - k.$

2

Show u five $u_5 = (-1)^4 u_4 + k = (7 - k) + k = 7.$

3

Use the given sum $\sum_{r=1}^4 u_r = 7 + (k - 7) + (2k - 7) + (7 - k) = 2k.$

4

Find kSince $\sum_{r=1}^4 u_r = 30$, we have $2k = 30$, so $k = 15$.

5

Write one repeated blockWith $k = 15$, the first four terms are 7, 8, 23, -8, and their sum is 30. Since $u_5 = u_1$, the pattern repeats every four terms.

6

Sum 150 termsThere are 37 complete blocks in the first 150 terms, with the first two terms left over. Hence $\sum_{r=1}^{150} u_r = 37(30) + 7 + 8 = 1125$.**FINAL ANSWER**

WMA12-01_2024_Oct_Q02

(a) Proven. (b) $k = 15$. (c) 1125.